optimisation, allowing engineers to monitor, predict, and enhance system behaviour without physical testing. This approach reduces downtime and boosts efficiency, particularly in manufacturing and logistics. Connectivity is vital for digitalisation and decarbonisation, allowing IoT devices like thermostats, EV chargers, and solar panels to intelligently manage energy consumption, optimising efficiency, and lowering carbon emissions. For instance, smart homes can adjust energy use based on real-time data, contributing to sustainability and energy savings.

Wi-Fi sensing: The next frontier

One of the most exciting developments in Wi-Fi technology is the rise of Wi-Fi sensing. This technology can detect movement and occupancy within a home, opening the door to new levels of contextual awareness. Imagine a home where the lighting and air-conditioning adjust automatically based on which room you are in, or a system that knows when you have left and powers down non-essential devices to save energy.

Wi-Fi sensing could also enhance home security by detecting unexpected movement when the home is unoccupied. This technology's potential spans improving energy efficiency to enabling seamless, personalised experiences. As Wi-Fi sensing evolves, it is set to become integral to the smart home and IoT landscape.

Wi-Fi 6 and the upcoming Wi-Fi 7 represent significant advancements in wireless networking, aiming to address modern connectivity needs in increasingly congested environments. Wi-Fi has evolved from prioritising speed to focusing on efficiency in handling multiple connected devices. With Wi-Fi 6, improving high device density and reducing power consumption is key, especially for IoT-heavy environments. This ensures stable connectivity for smart devices like sensors and smart locks while extending battery life. Wi-Fi 7 further enhances performance by reducing latency, which is critical for applications like virtual reality and advanced IoT systems.

Wi-Fi 6: Enhancing network efficiency, not just speed

Wi-Fi 6, also known as 802.11ax, was designed to improve efficiency, throughput, and performance in dense environments like homes with many smart devices, offices, or public venues. It builds upon the foundations of previous Wi-Fi generations, focusing on solving the shortcomings of earlier standards. The key features include:

Multi-user enhancements with OFDMA and MU-MIMO: Wi-Fi 6 routers utilise orthogonal frequency division multiple access (OFDMA) and multi-user, multiple input, multiple output (MU-MIMO) to handle multiple devices efficiently. OFDMA divides channels into smaller sub-channels, allowing better bandwidth allocation and lower latency, especially in crowded networks. MU-MIMO enables simultaneous communication with multiple devices, thus improving network efficiency. These technologies are crucial for today's environments with numerous IoT devices, video conferencing, and streaming. Devices fully supporting OFDMA and MU-MIMO perform better, particularly in congested networks.

Improved battery efficiency with target wake time (TWT): TWT allows devices to schedule when they wake up to send or receive data, reducing power usage and improving battery life. This is especially useful for IoT devices, wearables, and smartphones in smart homes, helping them stay connected while conserving energy. TWT is key in extending battery life without sacrificing connectivity in today's IoT environments.

Widespread 160MHz channel adoption: Wi-Fi 6 supports 160MHz channels, which double the amount of bandwidth compared to the 80MHz channels used in previous standards. This higher channel width translates to faster data transfer rates, which is especially useful for applications that require large bandwidth, such as 4K video streaming, VR, and online gaming. As more devices start to adopt 160MHz channel support, maximum throughput rates on Wi-Fi 6 networks are increasing. This trend is especially noticeable in high-end routers and mesh networking systems that cater to power users.

Wi-Fi 6E: Expanding into the 6GHz spectrum: The release of Wi-Fi 6E, an extension of Wi-Fi 6, brings the use of the 6GHz frequency band to Wi-Fi networks for the first time. This additional spectrum provides 1200MHz bandwidth, offering more non-overlapping channels and reducing congestion in heavily trafficked 2.4GHz and 5GHz bands. With Wi-Fi 6E, users experience less interference, higher throughput, and improved performance in areas with crowded 2.4GHz and 5GHz bands. As more devices become compatible with Wi-Fi 6E, it is expected to unlock the full potential of high-bandwidth applications, such as AR/VR, cloud gaming, and 8K video streaming.

Wi-Fi 7: Low latency and multi-link operation (MLO)

Wi-Fi 7, the upcoming iteration, builds upon many of the improvements in Wi-Fi 6 and focuses on latency-sensitive applications, such as gaming, virtual reality (VR), and edge computing. Wi-Fi 7 will introduce 320MHz channel widths, double that of Wi-Fi 6. Depending on the implementation and environment, this will allow for faster data transfer rates, potentially up to 46Gbps. The larger channel width will benefit bandwidth-hungry applications like cloud gaming, 8K video streaming, and real-time VR applications.

Multi-link operation (MLO): Wi-Fi 7 introduces MLO, a groundbreaking feature that allows devices to connect on multiple frequency bands (e.g., 2.4GHz and 5GHz) simultaneously. This is